Particle Physics 2: The Standard Model
Lecture 10: The Higgs Scale, Running Couplings, and the Hierarchy Problem

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Chapter 1

The Higgs Scale, Running Couplings, and the Hierarchy Problem

Overview. The final lecture begins by looking beyond the Standard Model toward unification, supersymmetry, and the gauge-hierarchy problem. To see why the Higgs field sits at the center of those questions, we first reconstruct the Dirac Lagrangian and show that a fermion mass mixes left and right chirality. The chiral weak interaction forbids that mass as a bare term; a gauge-invariant Yukawa coupling supplies it only after the Higgs field acquires a vacuum value. We then derive that vacuum value from the Higgs potential, ask why every known elementary mass is tied to this one small scale, and use superconductivity as a physical analog. After the break, the lecture develops renormalization from conductor and dielectric screening, carries the argument into QED and QCD, follows the gauge couplings toward unification, and derives the Planck scale from short-distance gravity. The electroweak scale then returns as the one unexplained small number.

1.1 The Question Waiting Beyond the Standard Model

The Standard Model is not presented here as a closed catalogue. At the start of the evening I want to say where the next course will go. The theory has difficulties rather than logical paradoxes: the gauge hierarchy, the possible role of supersymmetry, and the question of unification. The three gauge factors

$$SU(3)_c \times SU(2)_L \times U(1)_Y \tag{1.1}$$

describe strong, weak, and hypercharge interactions separately. Could they be subgroups of one larger group? An often studied possibility is $SU(5)$, in which quarks and leptons occupy common multiplets. Whether nature uses that particular construction is a different question. The important clue is that the apparently separate structures fit remarkably neatly into a larger mathematical one.

The Large Hadron Collider was built in part to expose whatever physics settles these questions. In 2010 its most immediate quarry was the Higgs boson. Finding a Higgs could never mean only adding one particle to a table. A Higgs is identified by an interlocking set of properties: its vacuum value gives masses to the weak vector bosons and fermions, while its fluctuation couples to those same particles with definite strengths. A resonance with the wrong couplings would not be the Standard Model Higgs. [1]

[1] Editorial clarification: The lecture predates the 2012 discovery. The observed approximately 125 GeV scalar has,

The route into that argument begins not with the Higgs potential but with the Dirac equation. We need to understand exactly what a fermion mass does before asking where it comes from.

1.2 From the Dirac Equation to Its Lagrangian

1.2.1 The older Dirac notation

In the Hamiltonian notation used when the Dirac equation was first introduced, its structure is

$$i\partial_t \psi = \left(-i\alpha^i \partial_i + \beta m\right)\psi. \tag{1.2}$$

Signs and factors of i depend on conventions, and the live board temporarily suppresses some of them. The invariant content is the separation between terms containing derivatives and a mass term containing none.

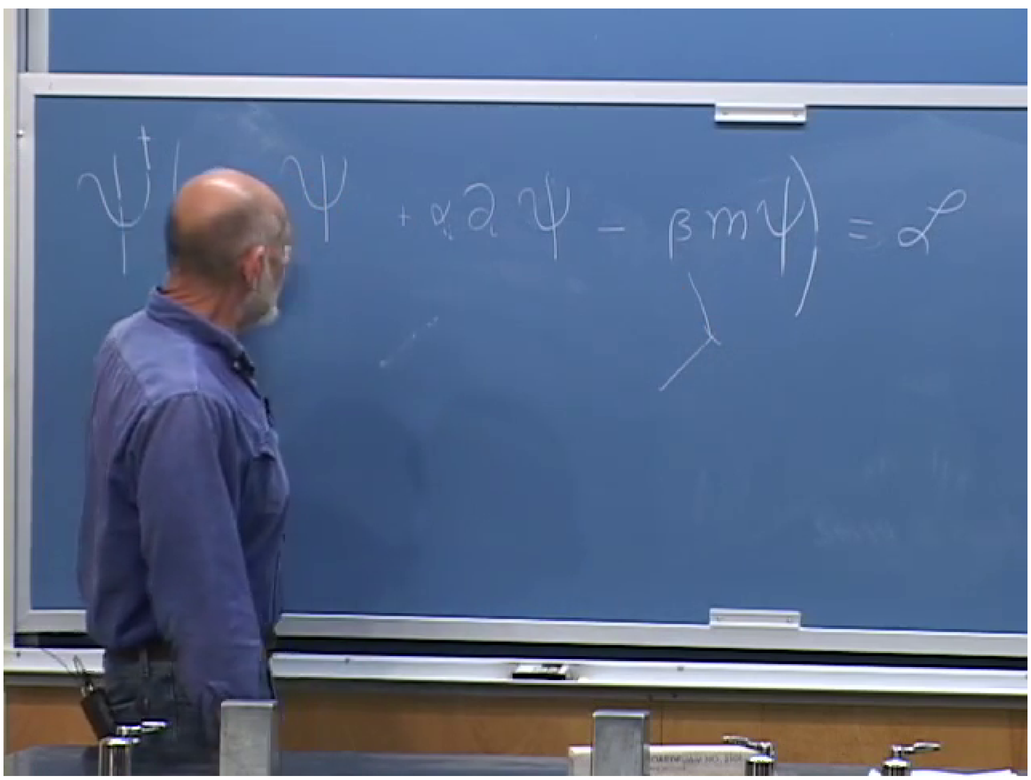

Figure 1.1: The Dirac equation and Lagrangian in the older α, β notation. The board is being used structurally rather than as a fixed sign convention.

Lecture frame: 00:05:52.

To obtain an action whose variation gives the Dirac equation, multiply the equation by $\psi^\dagger$. Moving every term to one side removes the need for an equal sign, prompting the evening's small mathematical joke: once an equation has been written as an expression that vanishes, the equal sign has done its

within present precision, the spin, parity, and coupling pattern expected of the Standard Model Higgs. No low-energy supersymmetric particles have been observed as of 2026. These later facts update the historical setting without changing the argument developed in the room.

work. The corresponding first-order Lagrangian may be written schematically as

$$\mathcal{L} = \psi^\dagger \left(i\partial_t + i\alpha^i \partial_i - \beta m \right) \psi, \tag{1.3}$$

with the understanding that a consistent metric convention fixes the relative signs.

The derivative terms move an excitation from one spacetime point to a nearby one and rotate its spinor components in the process. In a lattice picture they are hopping terms. By contrast, the mass term absorbs and re-emits the fermion at the same point. It has no derivative. This local operation will shortly reveal itself as a repeated conversion between two kinds of handedness.

Question from the audience. Should a γ^0 be placed after $\psi^\dagger$?[a]

Response. Not in the noncovariant form just written. But the question points directly to the covariant notation. Define the Dirac adjoint by $\bar{\psi} = \psi^\dagger \gamma^0$; then the γ^0 is indeed built into the object that stands on the left.

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[a] Lecture timestamp: 00:05:59.

1.2.2 The covariant form

Set

$$\bar{\psi} \equiv \psi^\dagger \beta = \psi^\dagger \gamma^0, \qquad \gamma^0 \equiv \beta, \qquad \gamma^i \equiv \beta \alpha^i. \tag{1.4}$$

Because $\beta^2 = 1$, multiplication by β converts freely between the two notations. The time derivative becomes $\bar{\psi}\gamma^0 \partial_0 \psi$, the spatial derivatives become $\bar{\psi}\gamma^i \partial_i \psi$, and the result is

$$\boxed{\mathcal{L}_D = \bar{\psi} \left(i\gamma^\mu \partial_\mu - m \right) \psi.} \tag{1.5}$$

This notation exposes the Lorentz character of each bilinear. $\bar{\psi}\psi$ is a scalar, whereas $\bar{\psi}\gamma^\mu\psi$ is a four-vector. The compact expression is not merely prettier than the α, β form; it tells us how the pieces transform.

Question from the audience. Was the spinor at the far left of the compact expression meant to be $\bar{\psi}$?[a]

Response. Yes. The missing bar is restored on the board. It matters: $\bar{\psi}$ is not $\psi^\dagger$, but $\psi^\dagger\gamma^0$.

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[a] Lecture timestamp: 00:09:39.

1.3 Chirality and What a Dirac Mass Actually Does

Besides $\gamma^0, \gamma^1, \gamma^2, \gamma^3$, define

$$\gamma^5 = i\gamma^0\gamma^1\gamma^2\gamma^3, \qquad (\gamma^5)^2 = 1. \tag{1.6}$$

The factor of i shown here is the standard convention for the mostly minus Minkowski metric. The eigenvalues are $+1$ and -1, and the corresponding projectors are

$$P_L = \frac{1-\gamma^5}{2}, \qquad P_R = \frac{1+\gamma^5}{2}, \qquad \psi_L = P_L\psi, \quad \psi_R = P_R\psi. \tag{1.7}$$

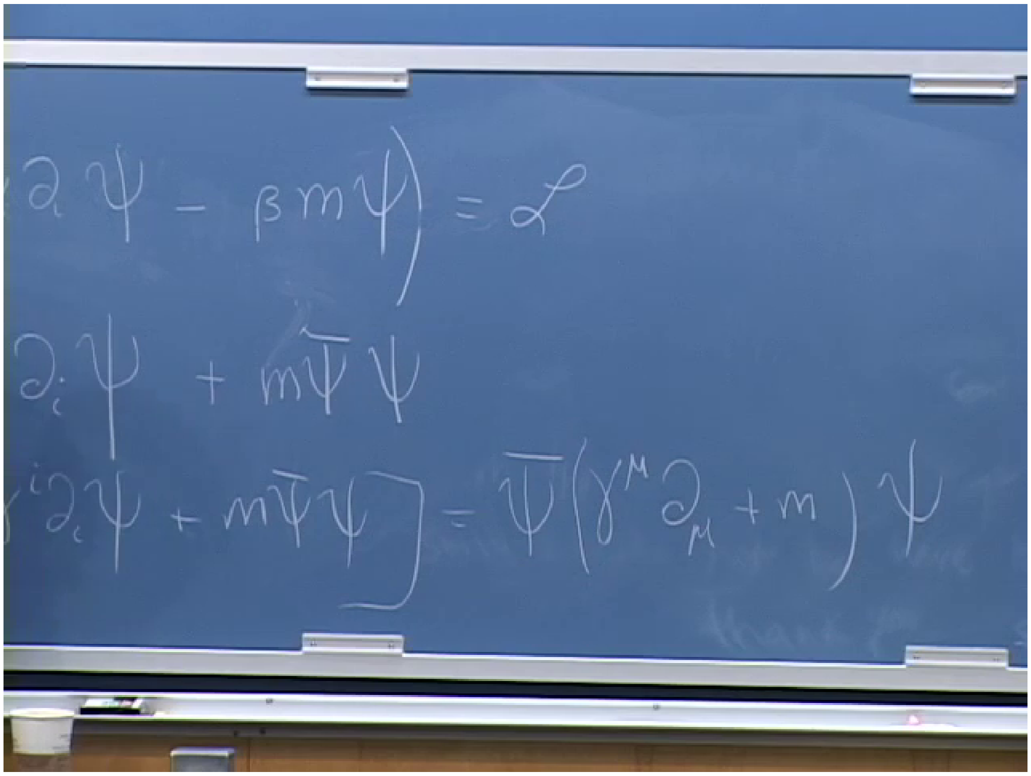

Figure 1.2: The covariant rewrite: $\bar{\psi} = \psi^\dagger\gamma^0$, $\gamma^i = \beta\alpha^i$, and the compact Dirac Lagrangian.
Lecture frame: 00:10:17.

Question from the audience. Why is the extra matrix called γ^5 when there is no γ^4? And does its eigenvalue describe helicity?[a]

Response. The name is historical: it is the additional matrix formed from the product of the four spacetime gamma matrices. Its eigenvalue labels chirality, not generally helicity. For a massless particle the two notions can be identified appropriately, which is why the language is easy to blur, but chirality is the Lorentz-representation label needed by the weak gauge theory.

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[a] Lecture timestamp: 00:10:42.

The kinetic term preserves chirality. One useful way to see this is that $\{\gamma^\mu, \gamma^5\} = 0$, so it separates into

$$\bar{\psi}i\gamma^\mu\partial_\mu\psi = \bar{\psi}_L i\gamma^\mu\partial_\mu\psi_L + \bar{\psi}_R i\gamma^\mu\partial_\mu\psi_R. \tag{1.8}$$

The scalar bilinear behaves differently:

$$\bar{\psi}\psi = \bar{\psi}_L\psi_R + \bar{\psi}_R\psi_L. \tag{1.9}$$

Thus a Dirac mass is a local chirality-flipping operation:

$$\psi_L \longleftrightarrow \psi_R. \tag{1.10}$$

A massless fermion may propagate with definite chirality. A massive one cannot remain in one chiral sector, because the mass term continually mixes the two. It may seem surprising that this conversion has anything to do with inertia, but it is precisely the relativistic field-theory form of fermion rest mass.

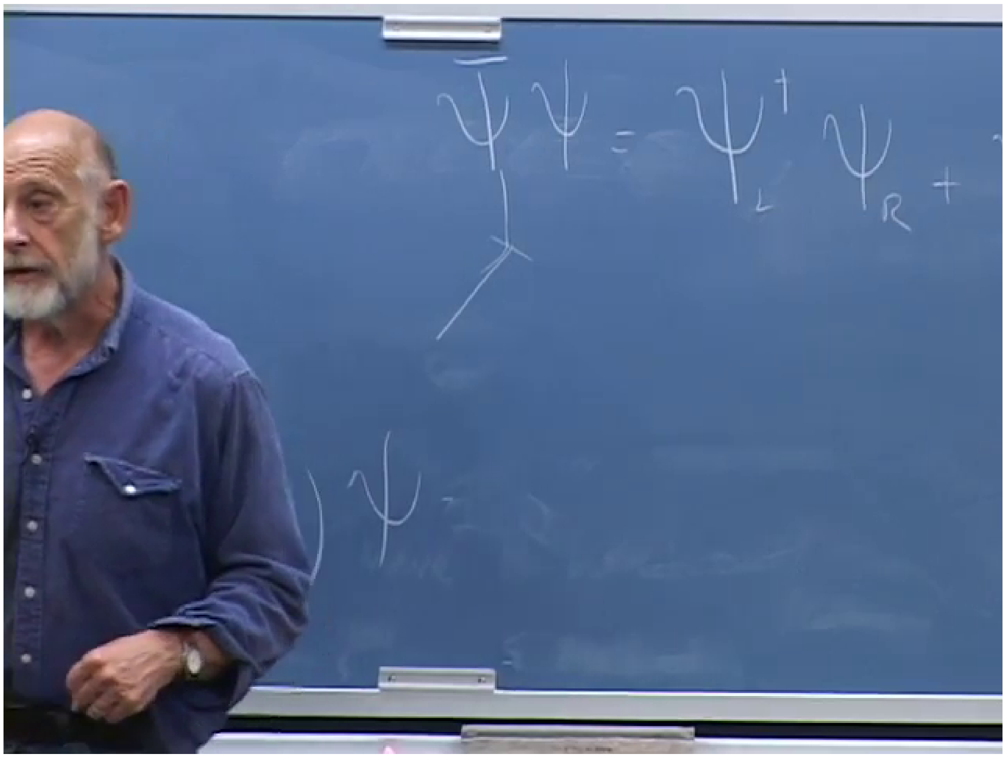

Figure 1.3: The board decomposes the scalar bilinear into left-to-right and right-to-left pieces.
Lecture frame: 00:15:38.

1.4 Why the Weak Symmetry Forbids a Bare Fermion Mass

The charged weak current is purely left-chiral. The lecture first names the Z, then corrects the statement to the $W^\pm$: Z couplings are chiral and unequal, but not simply absent for every right-chiral fermion. The clean statement is

$$\mathcal{L}_{\rm cc} = -\frac{g}{\sqrt{2}} \left(\bar{u}_L \gamma^\mu d_L + \bar{\nu}_L \gamma^\mu e_L\right) W_\mu^+ + \text{h.c.} \tag{1.11}$$

The left-handed fermions transform as $SU(2)_L$ doublets; the right-handed charged fermions transform as singlets.

Now compare those transformation rules with (1.9). A term such as $\bar{e}_L e_R$ absorbs an object carrying one set of electroweak quantum numbers and emits one carrying another. By itself it is not gauge invariant. The familiar Dirac mass is therefore illegal as a bare term in the electroweak Lagrangian.

This is the right order of explanation. Fermion masses are not first inserted and then cosmetically associated with the Higgs. Gauge symmetry first removes the obvious mass term. The Higgs field is then forced to carry the missing quantum numbers.

1.4.1 The Higgs as the charge carrier

Let the Higgs transform as an $SU(2)_L$ doublet with hypercharge $+\frac{1}{2}$:

$$H = \begin{pmatrix} H^+ \\ H^0 \end{pmatrix}, \qquad \widetilde{H} = i\sigma^2 H^*. \tag{1.12}$$

The lecture keeps the group indices schematic and calls the field ϕ. In that notation, a Yukawa term has the form

$$\mathcal{L}_Y = -G_Y \left(\bar{\psi}_L \phi \psi_R + \bar{\psi}_R \phi^\dagger \psi_L \right). \tag{1.13}$$

The Higgs field carries away the weak charge that a naked chirality flip would fail to conserve.

The complete one-generation Standard Model expression makes the same bookkeeping explicit:

$$\mathcal{L}_Y = -y_d \bar{Q}_L H d_R - y_u \bar{Q}_L \widetilde{H} u_R - y_e \bar{L}_L H e_R + \text{h.c.} \tag{1.14}$$

Each term is gauge invariant, and each fermion species carries its own dimensionless Yukawa coupling.

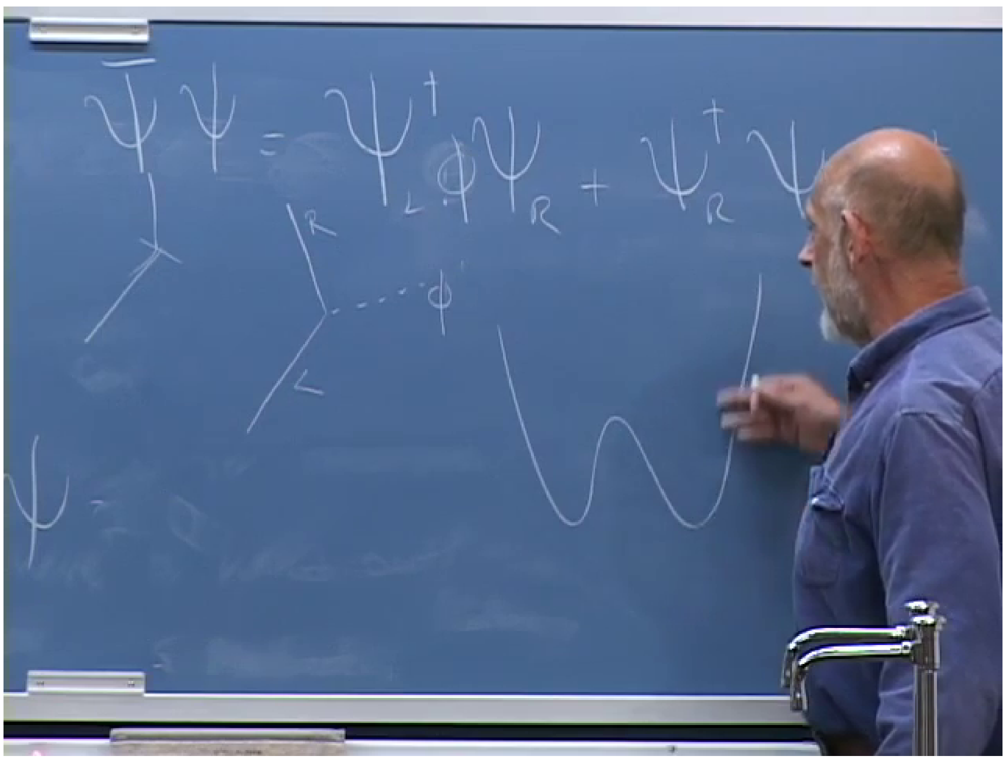

Figure 1.4: The Higgs-assisted chirality flip and the symmetry-breaking potential appear together on the board.

Lecture frame: 00:18:54.

1.5 One Interaction Becomes Both Mass and Higgs Coupling

Nature chooses a vacuum in which the neutral Higgs component does not vanish. In the lecture's schematic real-field notation,

$$\phi = F + \mathcal{H}, \tag{1.15}$$

where F is the nonfluctuating vacuum shift and $\mathcal{H}$ is the propagating Higgs fluctuation. Substitution into (1.13) gives

$$\begin{aligned} \mathcal{L}_Y = & -G_Y F \left(\bar{\psi}_L \psi_R + \bar{\psi}_R \psi_L \right) \\ & - G_Y \mathcal{H} \left(\bar{\psi}_L \psi_R + \bar{\psi}_R \psi_L \right). \end{aligned} \tag{1.16}$$

The first line is a Dirac mass and the second is an interaction with a physical Higgs quantum:

$$m_f = G_Y F \qquad \text{in the lecture's normalization.} \tag{1.17}$$

In the conventional Standard Model normalization,

$$H(x) = \frac{1}{\sqrt{2}} \begin{pmatrix} 0 \\ v + h(x) \end{pmatrix}, \qquad m_f = \frac{y_f v}{\sqrt{2}}, \qquad \mathcal{L}_{hff} = -\frac{m_f}{v} h \bar{f} f, \tag{1.18}$$

with $v \simeq 246\,\text{GeV}$. The lecture's rough $F \sim 200\,\text{GeV}$ is this same electroweak scale with normalization details suppressed.

The measured W and Z masses determine the scale F. The measured fermion masses then determine the Yukawa couplings. The electron's coupling is tiny; the top quark's is of order unity. If this were all the mechanism did, we would have put in one unknown number per fermion and read the same information back out.

But the second line of (1.16) is new experimental content. It says that a fermion can emit or absorb a Higgs and, read in the crossed channel, that a Higgs can decay into a fermion-antifermion pair. Schematically,

$$\mathcal{M}(h \to f\bar{f}) \propto y_f, \qquad \Gamma(h \to f\bar{f}) = N_c \frac{y_f^2 m_h}{16\pi} \left(1 - \frac{4m_f^2}{m_h^2}\right)^{3/2}, \tag{1.19}$$

at tree level. Here $N_c = 3$ for quarks and 1 for leptons. A larger fermion mass means a larger Yukawa coupling, provided the decay is kinematically allowed. This does not mean every heavy channel dominates: color factors, phase space, loop-induced channels, and decays to vector bosons also matter. It means the fermionic coupling itself is proportional to mass.

Question from the audience. How would one actually predict a Higgs decay time without doing every technical detail?[a]

Response. The interaction supplies a Hamiltonian matrix element. Draw the corresponding Feynman diagram, calculate its transition amplitude, square the amplitude, and sum or integrate over all allowed final states. The result is the decay rate. The lifetime is the inverse total width, $\tau = \hbar/\Gamma_{\text{tot}}$. This is the same quantum mechanical logic used in particle calculations since the early development of quantum field theory.

[a]Lecture timestamp: 00:25:32.

At the time of the lecture, this Higgs phenomenology was still prediction rather than direct observation. The W and Z self-interactions had already provided stringent tests of the gauge structure; the Higgs and its direct couplings remained to be found. The historical uncertainty belongs in the record. So does the later outcome: a Higgs-like particle was found, and measurements of its couplings are now tests of the same mass-generating structure developed here.

1.6 The Potential That Chooses F

The vacuum shift is not chosen by hand after writing the Yukawa coupling. It is determined by the Higgs potential. For one real cross-section through the full doublet, take

$$V(\phi) = -\frac{\mu^2}{2}\phi^2 + \frac{\lambda}{4}\phi^4, \qquad \mu^2 > 0, \quad \lambda > 0. \tag{1.20}$$

Near $\phi = 0$, the negative quadratic term makes an upside-down parabola. Calling μ an imaginary mass is shorthand for saying that the origin is unstable; it does not mean that the physical Higgs excitation about the true vacuum has an imaginary mass.

The quadratic term alone would drive the field without bound and leave no lowest-energy state. The positive quartic term eventually dominates and turns the potential upward. An odd power would spoil the toy $\phi \mapsto -\phi$ symmetry. A quartic is the first even stabilizing interaction and, in four spacetime dimensions, the highest polynomial scalar interaction with a dimensionless coupling.

Differentiate:

$$\begin{aligned} \frac{dV}{d\phi} &= -\mu^2\phi + \lambda\phi^3 \\ &= \phi\left(-\mu^2 + \lambda\phi^2\right). \end{aligned} \tag{1.21}$$

The stationary point $\phi = 0$ is the unstable top. The two minima satisfy

$$\boxed{F^2 = \frac{\mu^2}{\lambda}}, \qquad F = \frac{\mu}{\sqrt{\lambda}} \tag{1.22}$$

for this real-field convention.

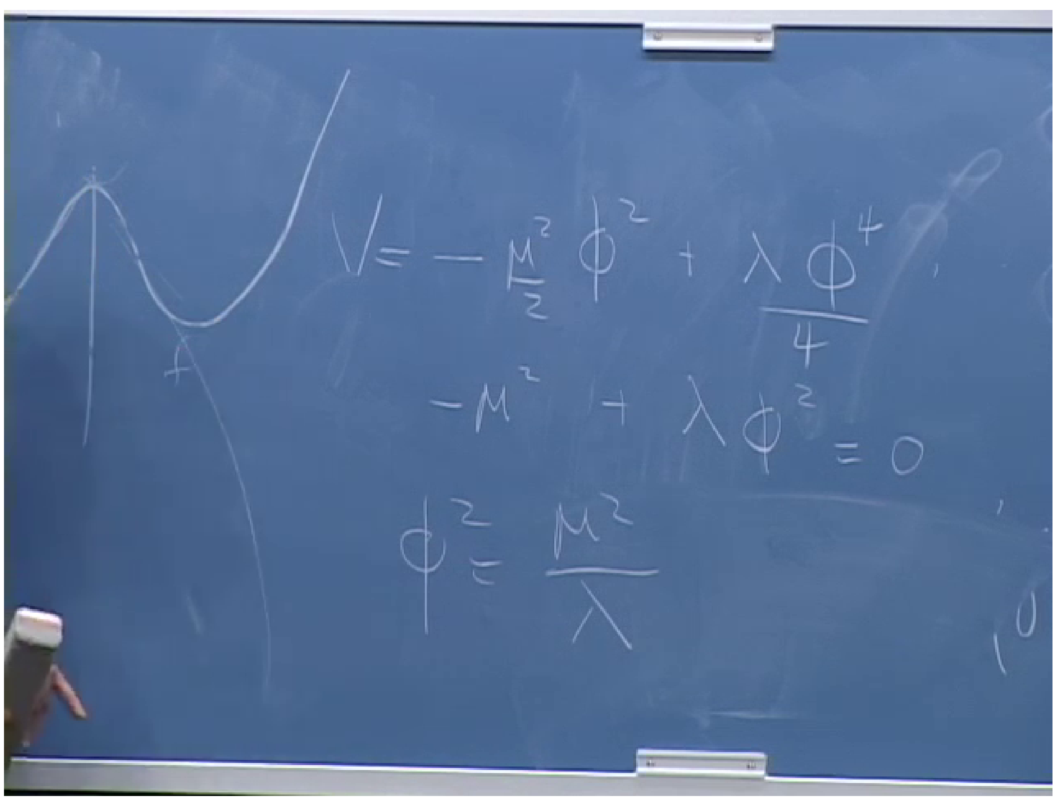

Figure 1.5: The completed minimization on the board: $V = -\mu^2\phi^2/2 + \lambda\phi^4/4$, $V' = 0$, and $\phi^2 = \mu^2/\lambda$.

Lecture frame: 00:34:32.

The factors 1/2 and 1/4 were inserted precisely so they disappear upon differentiation. Near the origin ϕ^4 is smaller than ϕ^2; at large field it grows faster and stabilizes the system. That simple competition produces the double well.

The dimensions also settle a momentary verbal slip in the transcript. λ is dimensionless in four spacetime dimensions, while μ and F have dimensions of mass. Expanding $\phi = F + \mathcal{H}$, the curvature of the potential gives

$$m_{\mathcal{H}}^2 = \left.\frac{d^2V}{d\phi^2}\right|_{\phi=F} = -\mu^2 + 3\lambda F^2 = 2\mu^2 = 2\lambda F^2. \tag{1.23}$$

In the full Standard Model notation, $V(H) = -\mu^2 H^\dagger H + \lambda(H^\dagger H)^2$, one likewise has $m_h^2 = 2\lambda v^2$. The measured Higgs mass and v imply λ near 0.13 at the electroweak scale. The lecture's 2010 estimate was intentionally rough; the structural claim was that λ is not an absurdly tiny number.

1.7 Why Is Everything We See Tied to This Scale?

The same F controls the W and Z masses and, multiplied by different Yukawa couplings, all charged-fermion masses. That empirical concentration raises a more interesting question than the algebra: why should all known elementary particles borrow their masses from the same symmetry-breaking phenomenon?

There is no theorem that every possible particle must. Imagine a vectorlike fermion whose left- and right-chiral components transform in the same way under the gauge group. Then $\bar{\psi}_L \psi_R + \text{h.c.}$ can be gauge invariant without a Higgs. A complete gauge singlet can likewise carry a mass unrelated to electroweak breaking. Such particles could naturally be much heavier than the particles tied to F. The particles we have discovered may be light enough to see precisely because symmetry would make them massless when F vanishes.

The electroweak scale is therefore the exceptional number. The speculative picture is that generic unprotected masses lie near a much larger fundamental scale, while particles protected by chiral gauge symmetry are dragged down with F.

Question from the audience. Is the Planck mass one candidate for the fundamental mass scale with which F should be compared?[a]

Response. Yes. Another is the scale suggested by gauge-coupling unification. Both will emerge from arguments later in the lecture, even though no particles of those enormous masses have been directly observed.

[a] Lecture timestamp: 00:38:43.

Question from the audience. What does it mean to say that the other scales are about fifteen orders of magnitude above F, and what particles correspond to them?[a]

Response. The electroweak scale is of order 10^2 GeV, while unification is suggested near 10^{16} GeV and the Planck scale lies near 10^{19} GeV. We have not produced particles at those masses. We infer the scales indirectly: one from extrapolating measured gauge couplings and the other from combining quantum mechanics, relativity, and Newton's constant.

[a] Lecture timestamp: 00:39:54.

1.8 A Charged Condensate in the Laboratory

Before the break, an audience question asks for an ordinary condensed matter analog of the Higgs mechanism. The answer develops in stages rather than by analogy alone.

Question from the audience. Beyond ultracold Bose condensates, is there a common solid-state example of a Higgs-like mechanism?[a]

Response. Yes. A neutral bosonic field can acquire a nonzero expectation value in a Bose condensate or superfluid. If the condensed object is charged, the same basic phenomenon alters the gauge field. Superconductivity supplies the familiar example: charged Cooper pairs condense, and electromagnetic fields acquire a finite penetration length inside the material.

[a]Lecture timestamp: 00:40:50.

At molecular resolution a helium atom can be treated as one effective bosonic particle with a field Φ_{He}. Empty space corresponds to zero field; a superfluid phase has $\langle \Phi_{\mathrm{He}} \rangle \neq 0$ and spontaneously breaks the relevant global phase symmetry. Helium atoms are neutral, so this is not yet the gauge-field mass mechanism.

In a superconductor, electrons form weakly bound Cooper pairs. The pair field is schematically an electron bilinear,

$$\Delta(x) \sim \langle \psi_\downarrow(x) \psi_\uparrow(x) \rangle, \tag{1.24}$$

and carries electric charge $2e$. When Δ condenses, the electromagnetic gauge field responds exactly as a gauge field coupled to a charged order parameter must. Static magnetic fields decay over the London penetration depth:

$$B(x) \propto e^{-x/\lambda_L}. \tag{1.25}$$

The inverse penetration length plays the role of an effective photon mass inside the medium. It is not a vacuum photon mass; it is a property of the superconducting phase.

Question from the audience. Are the two electrons in a Cooper pair not separated, even lying on opposite sides of the Fermi surface?[a]

Response. They are not a tiny molecule localized at one point. A Cooper pair is extended and is naturally described in momentum space by correlated states near opposite momenta. For the symmetry argument, however, the pair acts as a single bosonic charged order parameter. Its condensation is what matters.

[a]Lecture timestamp: 00:43:13.

Historically this relationship is more than a teaching metaphor. Superconductivity and Anderson's analysis of gauge fields in a broken phase helped make the relativistic Higgs mechanism intelligible. The particle-physics construction was developed independently in relativistic field theory, but the shared mathematics is real.

1.9 Bare Parameters and Measured Parameters

After the break, we return to the Lagrangian. It contains parameters called masses and charges, but those input parameters are rarely identical to quantities measured without qualification. High-frequency modes interact with low-frequency modes and dress them. The systematic relation between parameters defined at one resolution and observables at another is renormalization.

Electric charge gives a picture that can be built without calculation. At large separation, Coulomb's law defines the observed charge:

$$V(r) \sim \frac{e_1 e_2}{4\pi r}, \qquad F(r) \sim \frac{e_1 e_2}{4\pi r^2}. \tag{1.26}$$

The lecture suppresses 4π and writes e^2/r or e^2/r^2. What matters is that the coefficient inferred from a force measurement can depend on the distance at which the system is probed.

1.9.1 A conductor: complete screening

Place a positive test charge in an ideal conductor. Its electric field drives mobile negative charge toward it and positive charge away. Current flows until the local source is surrounded by an opposite cloud and no field remains in the bulk outside that cloud; compensating charge is pushed to the remote boundary. A negative source produces the opposite rearrangement. Two dressed sources far apart can then exert essentially no force, even though the charges inserted microscopically were nonzero.

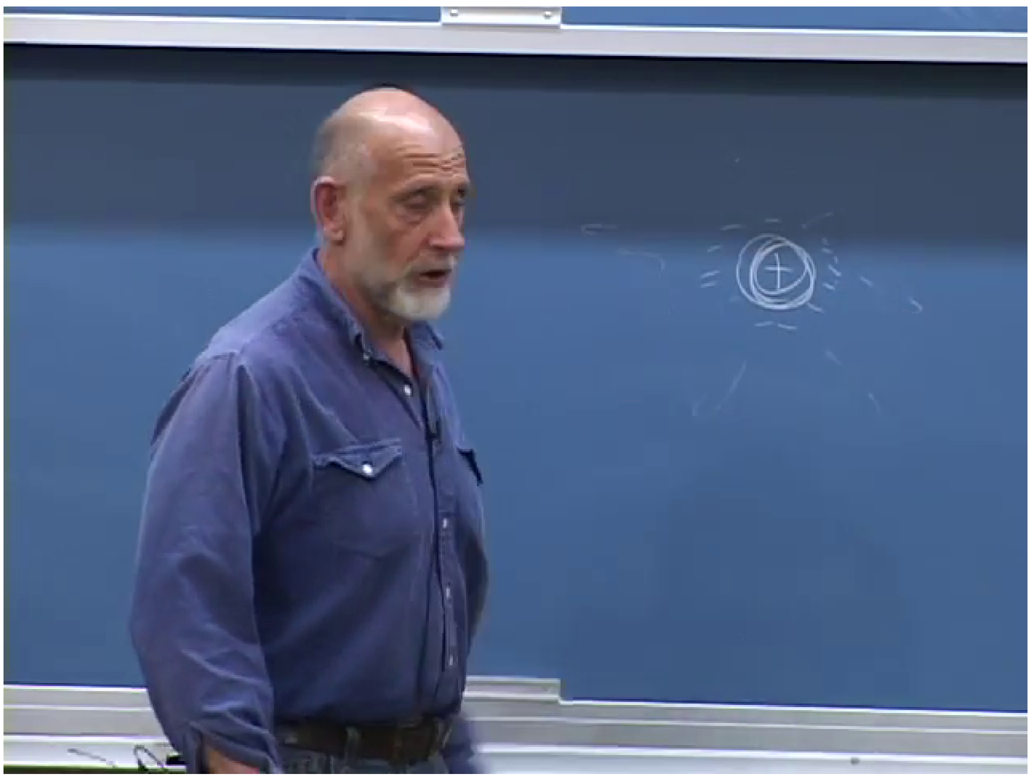

Figure 1.6: A source charge surrounded by an oppositely charged screening cloud in the conductor discussion.

Lecture frame: 00:50:24.

Question from the audience. Is this rearrangement what is meant by screening?[a]

Response. Yes. At distances larger than the cloud, the inserted charge is screened. Bring another probe inside the cloud and some of the screening can no longer lie between source and probe; the underlying charge then appears stronger.

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[a] Lecture timestamp: 00:51:25.

The cloud has a characteristic scale set by the dynamics and density of the mobile charges. At separations r smaller than that scale, the probe penetrates the cloud. One may therefore define a running charge $e(r)$ through

$$F(r) \equiv \frac{e^2(r)}{4\pi r^2}. \tag{1.27}$$

For the idealized conductor, $e(r)$ approaches zero at very long distance and the microscopic value at very short distance.

1.9.2 A dielectric: partial screening

In an insulator, electrons are bound to atoms rather than free to cross the sample. A crude but useful model pictures each electron cloud attached to its ion by a spring. Without an external field, the average positive and negative charge centers coincide. An applied field shifts them slightly apart and induces a dipole.

Around a negative source, nearby electron clouds shift away and leave their positive ions slightly closer; around a positive source they shift the other way. The bound charges cannot flow far enough to neutralize the source completely, so the long-distance field is reduced by the dielectric constant rather than eliminated:

$$V_{\text{medium}}(r) = \frac{1}{\epsilon_r}\frac{e_1 e_2}{4\pi\epsilon_0 r}. \tag{1.28}$$

At distances inside the polarization cloud, the probe again sees more of the bare source. Complete and partial screening are two versions of the same scale-dependent idea.

1.10 The QED Vacuum as a Polarizable Medium

Quantum electrodynamics has no material atoms in its vacuum, but it has charged quantum fields. Electron-positron fluctuations contribute loops to photon propagation. In the older Dirac-sea language, the electric field distorts the filled negative-energy states; in modern language, vacuum polarization dresses the photon propagator. The medium picture says that a virtual pair is polarized: its positive and negative components shift in opposite directions.

A positive source is surrounded by a small excess of negative polarization charge, and a negative source by positive polarization charge. The charge measured far away is therefore smaller than the parameter one would associate with an unresolved source. Unlike an ordinary material, the vacuum contains fluctuations over a continuum of momenta. Low-energy pairs correspond to longer distance structure; high-energy pairs probe shorter distances. There is no final cloud radius at which the process simply stops.

As two electrons are brought closer together, the largest polarization clouds cease to fit between them first. Bringing them closer peels away successively smaller clouds. The effective electric coupling grows logarithmically with inverse distance, or equivalently with energy:

$$\alpha(Q) \simeq \frac{\alpha(\mu)}{1 - \dfrac{2\alpha(\mu)}{3\pi}\ln\left(\dfrac{Q}{\mu}\right)} \tag{1.29}$$

for one charged Dirac fermion well above its threshold at leading order. Additional charged species change the coefficient when their thresholds are crossed.

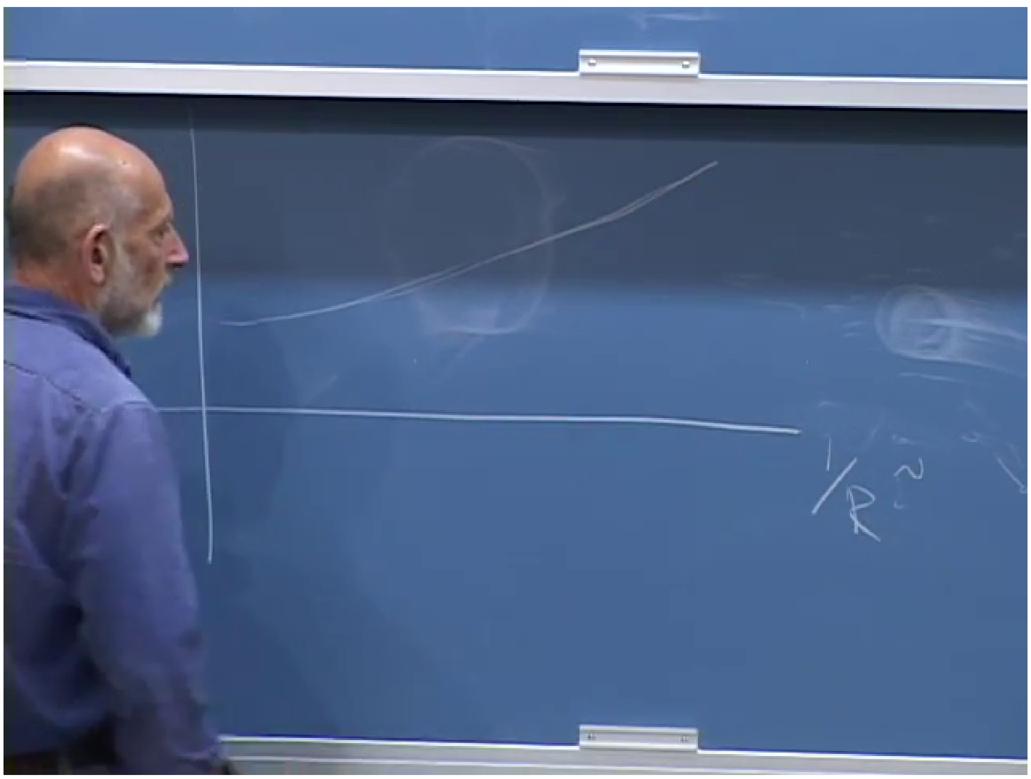

Figure 1.7: The QED effective charge rises slowly as the horizontal scale $1/R$, and therefore the probing energy, increases.

Lecture frame: 01:02:13.

Electron scattering gives the operational version. Higher collision energy means shorter de Broglie wavelength and smaller closest approach. The amplitude must therefore use the coupling appropriate to that momentum transfer, not one fixed number for every scale.

Question from the audience. At what distance does the running first become measurable?[a]

Response. Electron vacuum polarization becomes relevant around the electron Compton scale, $\lambda_C = \hbar/(m_e c)$, roughly 4×10^{-11} cm for the reduced wavelength. The effect is small and logarithmic, so precision is required. Heavier charged particles join the running at their own thresholds.

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[a]Lecture timestamp: 01:03:54.

Question from the audience. Does speaking of the smallest distance assume that we already know what happens there?[a]

Response. No. We extrapolate the theory only while its particle content and interactions remain valid. New physics can change the running. Experiment does not tell us that QED remains the complete description to arbitrarily short distance.

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[a]Lecture timestamp: 01:04:49.

Question from the audience. For gluons, is the scale dependence the opposite? Why? [a]

Response. Yes. Quark loops screen color charge in a way analogous to charged matter, but gluons themselves carry color and interact nonlinearly. Their contribution has the opposite sign and dominates for the observed number of quark flavors. The result is antiscreening and asymptotic freedom.

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[a]Lecture timestamp: 01:05:03.

Question from the audience. How could one determine a bare electric charge if doing so requires an infinitely small distance?[a]

Response. One does not measure a unique regulator-independent bare charge directly. One defines a parameter at a chosen short-distance cutoff or renormalization scale, then adjusts it so a measured low-energy quantity remains fixed. The scale-dependent renormalized coupling is the useful physical object.

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[a]Lecture timestamp: 01:05:28.

Question from the audience. Does the QED charge approach a finite value at infinite energy?[a]

Response. Not in perturbative QED extrapolated by itself. Its positive beta function makes the coupling continue to grow logarithmically and leads formally to a Landau pole at a fantastically high scale. Long before treating that formal pole as physical, one expects the description to be embedded in electroweak theory and ultimately altered by further physics.

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[a]Lecture timestamp: 01:05:39.

1.10.1 What renormalization changes and what it preserves

Choose a cutoff length a, call the corresponding input $e(a)$, and calculate a fixed long-distance observable. A different cutoff a' is equally acceptable if the input is changed to $e(a')$ so the same observable is reproduced:

$$\frac{d}{d\ln\mu}\,\mathcal{O}_{\text{physical}} = 0, \qquad \mu\frac{de}{d\mu} = \beta(e). \tag{1.30}$$

That compensating flow is the renormalization group.

Question from the audience. Must the word bare refer specifically to the Planck length?[a]

Response. No. One may terminate the description at any chosen cutoff and call the parameter there bare. Changing the cutoff while changing that parameter appropriately leaves accessible physics unchanged. The Planck length is a plausible place where ordinary quantum field theory must be replaced, not a required convention for renormalization.

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[a] Lecture timestamp: 01:06:20.

Question from the audience. Is this procedure a matter of zooming in and attenuating the description?[a]

Response. It is zooming in conceptually. We ask how the parameters of a finer-grained description must change so that predictions at a fixed, experimentally accessible wavelength remain the same.

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[a] Lecture timestamp: 01:07:15.

Question from the audience. Does the same running phenomenon happen with gravity?[a]

Response. Gravity has scale dependence too, but not simply the same dielectric-screening mechanism. For the present lecture a dimensional argument is enough: localization raises the energy, energy gravitates, and the resulting dimensionless gravitational strength grows rapidly with energy.

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[a] Lecture timestamp: 01:08:14.

1.11 QCD: Antiscreening, Flux Tubes, and Asymptotic Freedom

At hadronic distances, chromoelectric field lines do not spread spherically like an electromagnetic Coulomb field. Gluon self-interactions squeeze them into a narrow tube between a quark and antiquark. If the tube has approximately fixed energy per unit length σ, then

$$V_{q\bar{q}}(r) \simeq -\frac{C_F\alpha_s(r)}{r} + \sigma r. \tag{1.31}$$

At large r, the linear term dominates. Stretching the pair creates more of the tube, like pulling a strand of taffy, and the energy grows rather than tending to zero. This is the confinement side of the story.

At distances shorter than the tube radius, the potential returns approximately to a Coulomb form, but its coupling decreases with energy. At one loop,

$$\alpha_s(Q) = \frac{4\pi}{\left(11 - \frac{2}{3}n_f\right)\ln(Q^2/\Lambda^2_{\text{QCD}})}. \tag{1.32}$$

For $n_f < 17$, the coefficient is positive and $\alpha_s(Q) \to 0$ as $Q \to \infty$. QCD is weak at very short distance and strong at long distance. This is the precise running-coupling version of the lecture's opposite behavior.

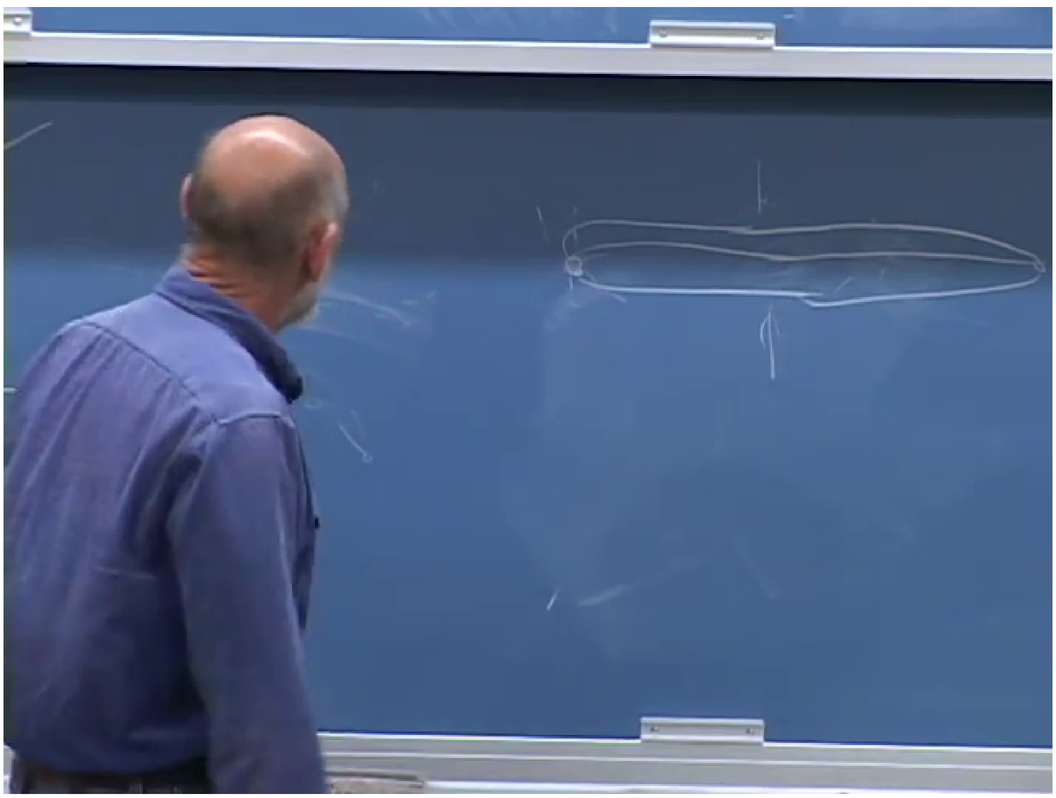

Figure 1.8: Chromoelectric field lines are drawn as a tube joining separated color sources rather than as a freely spreading Coulomb field.

Lecture frame: 01:09:56.

Question from the audience. What happens to the weak coupling between the QED and QCD curves?[a]

Response. Its running is intermediate and depends on which electroweak factor is being followed. Above the electroweak scale the proper quantities are the $U(1)_Y$ and $SU(2)_L$ couplings. One rises and the other falls slowly. Together with the falling strong coupling, their extrapolated values come surprisingly close at high energy.

[a]Lecture timestamp: 01:11:53.

1.12 Running Toward Unification

Measure the three gauge couplings near the weak scale and evolve them with the particle content of a proposed theory. Because every beta function is logarithmic, an enormous change in energy produces only a moderate change in coupling. The Standard Model trajectories approach one another but do not meet exactly. With the low-energy particle content of the minimal supersymmetric Standard Model, the one-loop trajectories meet much more closely near

$$M_{\mathrm{GUT}} \sim 10^{16}\,\mathrm{GeV}. \tag{1.33}$$

The plot is evidence, not proof. Thresholds from undiscovered particles, higher-loop corrections, and details of a unified theory shift the curves. The 2010 lecture emphasizes the approximately percent-level convergence in a supersymmetric spectrum because it made weak-scale supersymmetry especially attractive. The absence of superpartners at collider energies now means that the simplest version of that spectrum is not realized as originally expected; precision unification remains model dependent.

Why regard the vicinity of a meeting point as fundamental? It is where the input may become simplest. Three unrelated low-energy couplings could be different dressed descendants of one high-energy coupling associated with a single gauge group. This is the numerical clue behind the earlier $SU(5)$ question.

Gravity appears on the same logarithmic sketch as a curve that rises much faster. It does not cross the gauge curves at exactly the GUT scale, but it becomes order one within a few decades of energy. On a logarithmic axis, that proximity is suggestive: 10^{16}, 10^{17}, 10^{18}, and 10^{19} GeV are vastly closer to one another than any is to the electroweak scale.

1.13 Why Gravity Strengthens at Short Distance

1.13.1 Energy, not rest mass alone, gravitates

Newton's force between two slowly moving masses is

$$F_G = \frac{G m_1 m_2}{r^2}. \tag{1.34}$$

For a relativistic dimensional estimate, replace each rest mass by the corresponding energy divided by c^2:

$$F_G(r) \sim \frac{G E_1 E_2}{c^4 r^2}. \tag{1.35}$$

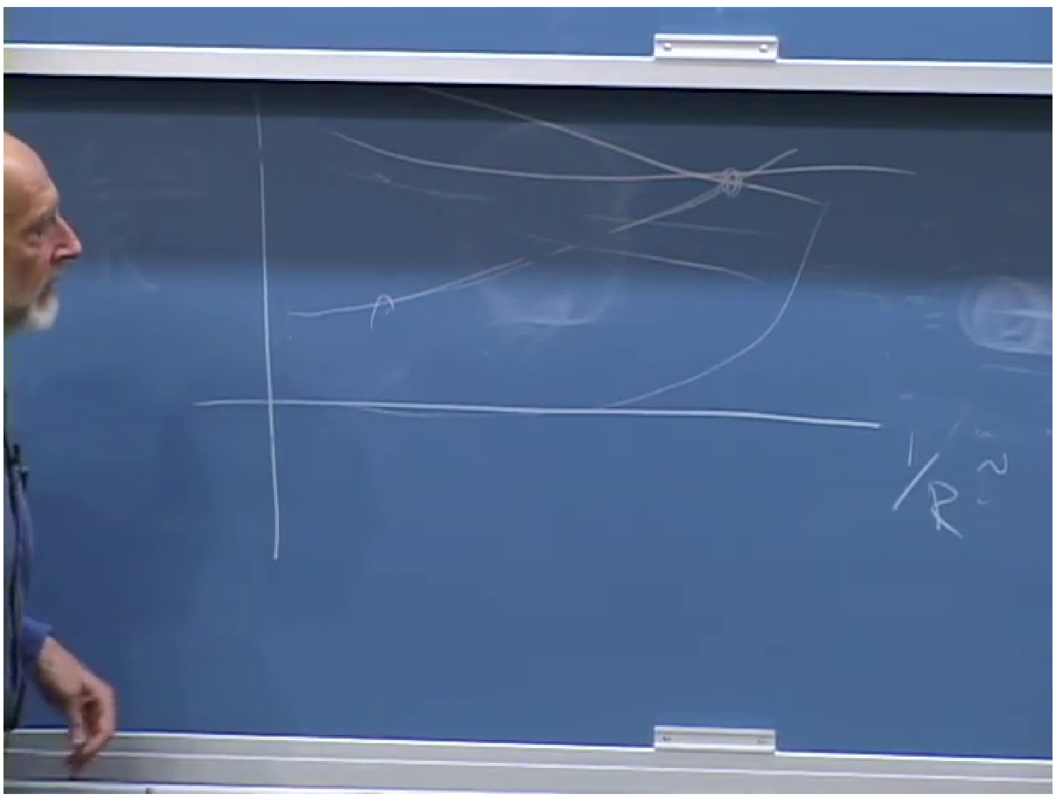

Figure 1.9: The board's qualitative coupling plot: the strong curve falls, the electroweak curves evolve slowly toward it, and gravity rises steeply near the far ultraviolet.

Lecture frame: 01:16:29.

This is not a substitute for general relativity; stress, momentum flow, and pressure also source gravity. It is the correct scaling estimate for the question at hand.

Localize each particle to a region of size r. The uncertainty principle requires

$$p \sim \frac{\hbar}{r}. \tag{1.36}$$

At sufficiently small r, the momentum is ultrarelativistic, so

$$E \simeq pc \sim \frac{\hbar c}{r}. \tag{1.37}$$

Substituting two such energies into (1.35) gives

$$F_G(r) \sim \frac{G\hbar^2}{c^2 r^4}. \tag{1.38}$$

The corresponding potential-energy scale is

$$U_G(r) \sim -\frac{G\hbar^2}{c^2 r^3}. \tag{1.39}$$

This is the distinction corrected during the live calculation: the potential scales as $1/r^3$, while the force scales as $1/r^4$.

Question from the audience. For an ultrarelativistic particle, is the energy $p^2/(2m)$?[a]

Response. No. That is the nonrelativistic formula. From $E^2 = p^2c^2 + m^2c^4$, the limit $pc \gg mc^2$ gives $E \simeq pc$. Combined with $p \sim \hbar/r$, the localization energy is $\hbar c/r$.

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[a]Lecture timestamp: 01:22:18.

Question from the audience. Did the factors of c cancel, and was the claimed $1/r^3$ law a force law?[a]

Response. The board calculation is corrected in real time. The two effective masses contribute E_1E_2/c^4; with $E_i \sim \hbar c/r$, two powers of c remain in the denominator. The resulting force is $G\hbar^2/(c^2r^4)$. The $1/r^3$ expression belongs to the potential energy.

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[a]Lecture timestamp: 01:23:38.

Question from the audience. The gravitational force under discussion is attractive, correct?[a]

Response. Yes. The equations above display magnitudes. The Newtonian gravitational potential carries a minus sign for positive energies, so the force is attractive in this estimate.

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[a]Lecture timestamp: 01:24:42.

Question from the audience. Is gravity gaining relative strength because localization raises energy, while electric charge itself does not gain the same kinematic factor?[a]

Response. Exactly. Energy is the gravitational charge. Squeezing a quantum state raises its momentum and energy, so it also raises what sources gravity. Electric charge is dimensionless and does not acquire that direct $1/r$ factor, apart from its much slower logarithmic renormalization.

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[a]Lecture timestamp: 01:25:02.

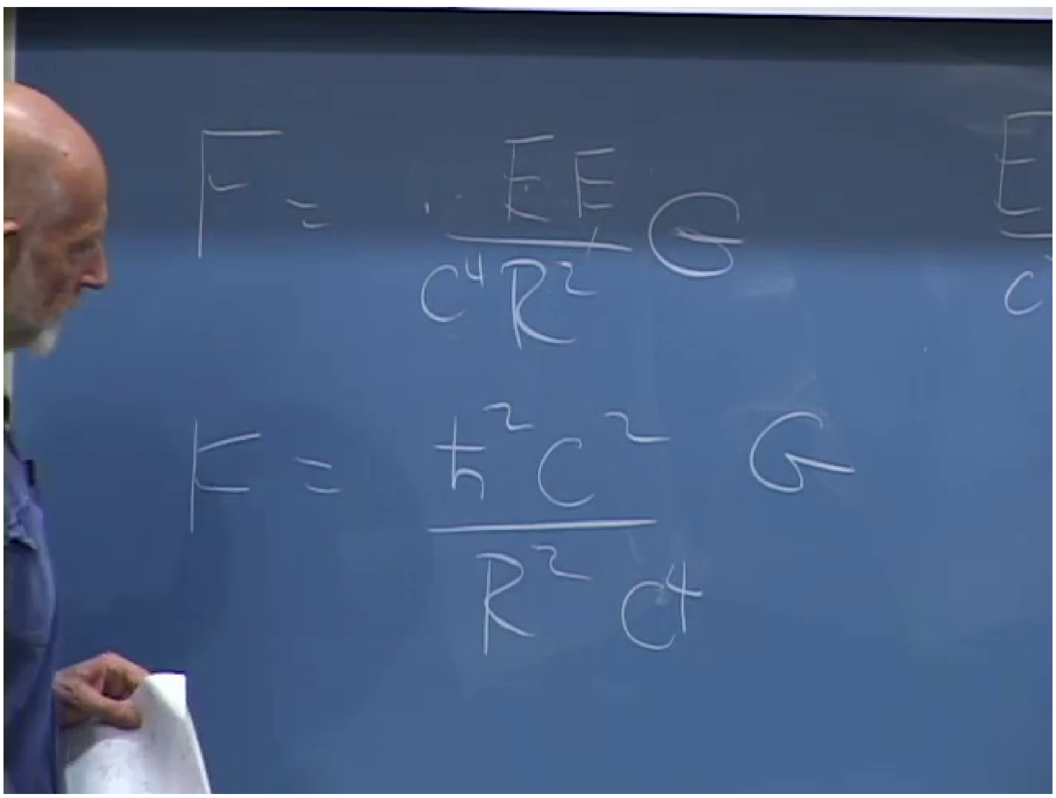

Figure 1.10: The board substitutes relativistic energies into Newton's law while checking the powers of c and r.

Lecture frame: 01:23:29.

1.13.2 The Planck crossing

The electromagnetic force between charges of order one in natural units has magnitude

$$F_{\rm EM}(r) \sim \alpha\,\frac{\hbar c}{r^2}. \tag{1.40}$$

Equating it to (1.38) gives

$$r^2 \sim \frac{G\hbar}{\alpha c^3}. \tag{1.41}$$

The lecture sets the dimensionless gauge factor to order one to expose the fundamental gravitational length:

$$\boxed{\ell_P = \sqrt{\frac{\hbar G}{c^3}}} \simeq 1.616 \times 10^{-35}\,\mathrm{m}. \tag{1.42}$$

Keeping the actual α moves the equality with electromagnetism by an order-one logarithmic-scale factor, $r \sim \ell_P/\sqrt{\alpha}$. The fundamental statement is cleaner in natural units:

$$\alpha_G(E) \equiv \frac{GE^2}{\hbar c^5} = \left(\frac{E}{E_P}\right)^2, \qquad E_P = \sqrt{\frac{\hbar c^5}{G}} \simeq 1.22 \times 10^{19}\,\mathrm{GeV}. \tag{1.43}$$

Gravity becomes strong when E approaches E_P.

Question from the audience. Was G or the electric coupling being set equal to one in the comparison?[a]

Response. The electric coupling was being treated as an order-one number for the rough estimate. Newton's constant must remain: it is the dimensional quantity that determines the Planck scale.

[a] Lecture timestamp: 01:27:02.

Question from the audience. The intermediate powers of r, $\hbar$, and c do not appear dimensionally consistent. What is the corrected calculation?[a]

Response. Use energies from the start. Localization gives $E \sim \hbar c/r$, hence $U_G \sim GE^2/(c^4 r) = G\hbar^2/(c^2 r^3)$. Compare this with $U_{\rm EM} \sim \alpha\hbar c/r$. Setting α to order one yields $r^2 = G\hbar/c^3$. The class and board arrive at $r = \sqrt{G\hbar/c^3}$, the Planck length.

[a] Lecture timestamp: 01:27:47.

1.14 The Hierarchy Problem Returns

The gauge couplings become comparable near a very high scale, and gravity becomes strong near the Planck scale. These enormous scales now look less like arbitrary large numbers and more like places where a microscopic description might simplify. Against them, the electroweak vacuum value F is anomalously small:

$$\frac{v}{E_P} \sim 2 \times 10^{-17} \tag{1.44}$$

in units with $c = 1$. Using a GUT scale instead of the Planck scale changes the precise ratio but not the puzzle.

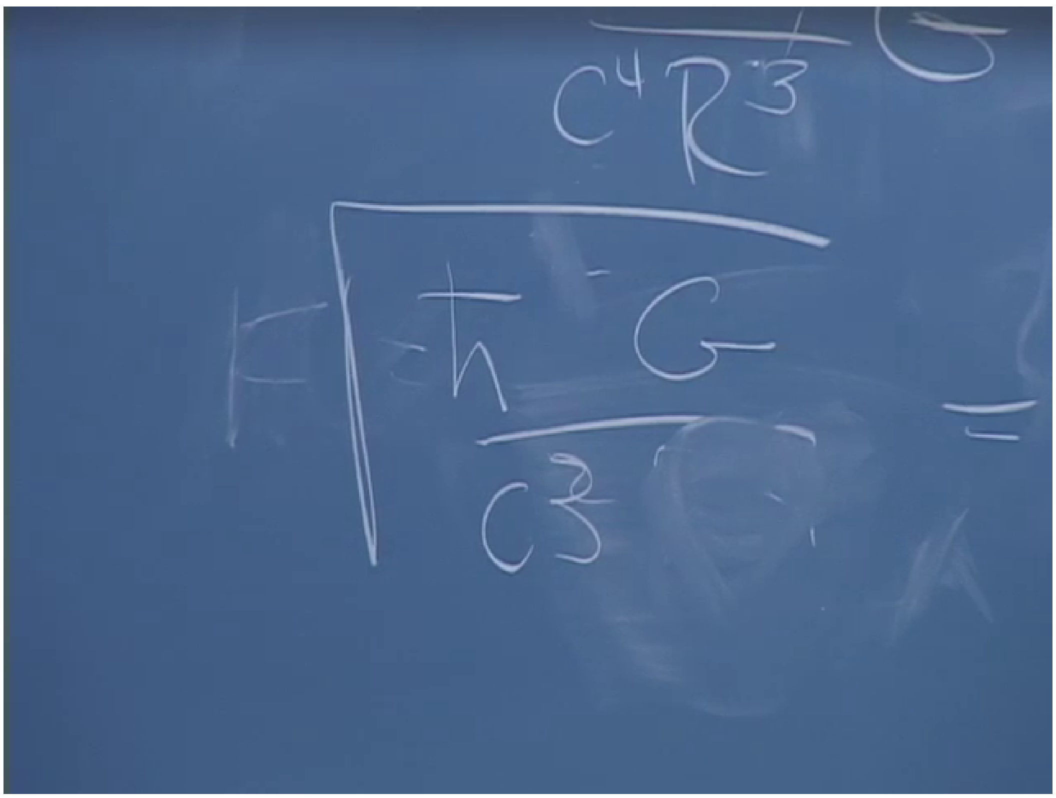

Figure 1.11: The corrected endpoint of the live dimensional check: $r = \sqrt{\hbar G/c^3}$.

Lecture frame: 01:32:16.

Once F is small, the masses of the W, Z, charged leptons, and quarks follow because gauge symmetry makes their masses proportional to it. Particles without that protection can remain heavy. The outstanding question is therefore not why every known mass is independently tiny. It is why the Higgs sector supplies this one tiny symmetry-breaking scale.

Question from the audience. If one could calculate F, would all the other masses then be calculable?[a]

Response. It would determine the overall electroweak scale and, with the gauge and Yukawa couplings supplied, the corresponding particle masses. The remaining problem is that an ordinary ultraviolet-sensitive calculation tends to drive the Higgs mass parameter toward the high fundamental scale unless a symmetry or dynamical mechanism protects it.

[a] Lecture timestamp: 01:34:39.

The word fine-tuned anticipates that sensitivity. In a cutoff description, a scalar mass parameter receives corrections of the rough form

$$\delta\mu^2 \sim \frac{C}{16\pi^2}\Lambda^2, \tag{1.45}$$

where C combines gauge, Yukawa, and scalar couplings and Λ marks the scale at which the effective theory is replaced. If Λ is near a GUT or Planck scale, obtaining $\mu^2 \sim v^2$ appears to require a very precise cancellation between the input and radiative contributions. This is the naturalness form of the gauge-hierarchy problem. It is an editorial completion of the calculation the lecture promises for the following quarter, not a claim that a particular regulator artifact is itself observable.

Question from the audience. Should the Planck length be visualized as the final screening-cloud radius?[a]

Response. No. The dielectric picture explains gauge-charge renormalization, but the Planck argument used localization energy and the dimensional gravitational coupling. There is no literal gravitational screening cloud in this derivation.

[a] Lecture timestamp: 01:35:03.

1.14.1 The closing questions about interactions

The last minutes return to the Higgs potential and to what terms a Lagrangian may contain.

Question from the audience. Could the Higgs potential include ϕ^6 and still remain symmetric?[a]

Response. Yes. Even powers preserve the toy $\phi \mapsto -\phi$ symmetry, whereas ϕ^5 would break it. In four dimensions the coefficient of ϕ^6 has negative mass dimension, so the term is suppressed by a high scale, for example $c_6\phi^6/\Lambda^2$. It is a valid effective interaction even though it is not power-counting renormalizable.

[a] Lecture timestamp: 01:35:20.

Question from the audience. Do the powers of the Higgs field represent creation and annihilation processes, so that ϕ^4 predicts Higgs-Higgs scattering?[a]

Response. Yes. After expanding about the vacuum, powers of the quantized field generate vertices. The quartic term contains a four-Higgs vertex, and the shifted potential also produces cubic interactions. Knowing their coefficients predicts processes such as $hh \to hh$, subject to the usual diagrammatic calculation.

[a] Lecture timestamp: 01:35:52.

Question from the audience. Must a fundamental Lagrangian be polynomial in its fields, with no power above four?[a]

Response. For a perturbatively renormalizable four-dimensional scalar theory, operators of dimension at most four form the traditional finite list. Modern effective field theory is less restrictive: higher dimension operators are allowed and encode short-distance physics, but their effects are suppressed by powers of the high scale. Thus the occasional exception is essential to the old rule.

[a]Lecture timestamp: 01:36:09.

Question from the audience. Were self-interaction terms omitted from the discussion of running couplings?[a]

Response. They were included conceptually. Gluon self-interactions are exactly what reverse the sign relative to QED and produce antiscreening. Higgs self-interactions likewise contribute to the running of scalar and other Standard Model parameters.

[a]Lecture timestamp: 01:36:44.

Question from the audience. Are there cross-terms in which different interactions affect one another's running?[a]

Response. Yes. At higher loop order, beta functions contain mixed products of gauge, Yukawa, and scalar couplings. Perturbation theory remains useful while those dimensionless couplings are sufficiently small, because successive loop orders are correspondingly suppressed.

[a]Lecture timestamp: 01:37:23.

Question from the audience. What exactly was meant by saying that F is finely tuned?[a]

Response. The complete answer is deferred in the lecture. The issue is the instability of a light scalar scale against much larger ultraviolet contributions, summarized by (1.45). Supersymmetry was one proposed way to enforce cancellations, but the hierarchy problem itself does not select a confirmed solution.

[a]Lecture timestamp: 01:37:44.

Question from the audience. Can the renormalization group itself be explained now? [a]

Response. Not in the remaining minutes. Its operative content has already appeared: change the resolution, let the couplings flow, and preserve fixed physical predictions. A full treatment belongs to the next stage of the course.

[a]Lecture timestamp: 01:37:56.

That is where the sequence properly leaves us. The Higgs mechanism has made the Standard Model internally coherent: it preserves the chiral gauge symmetry, gives masses to weak bosons and fermions, and predicts physical Higgs interactions. Renormalization then teaches us how those interactions change with scale and hints that gauge forces may share a unified origin. Gravity supplies a second high scale by an independent argument. None of those successes explains why the Higgs vacuum value is so far below both. The final lecture therefore does not close the subject; it identifies the small number on which the next subject must concentrate.